

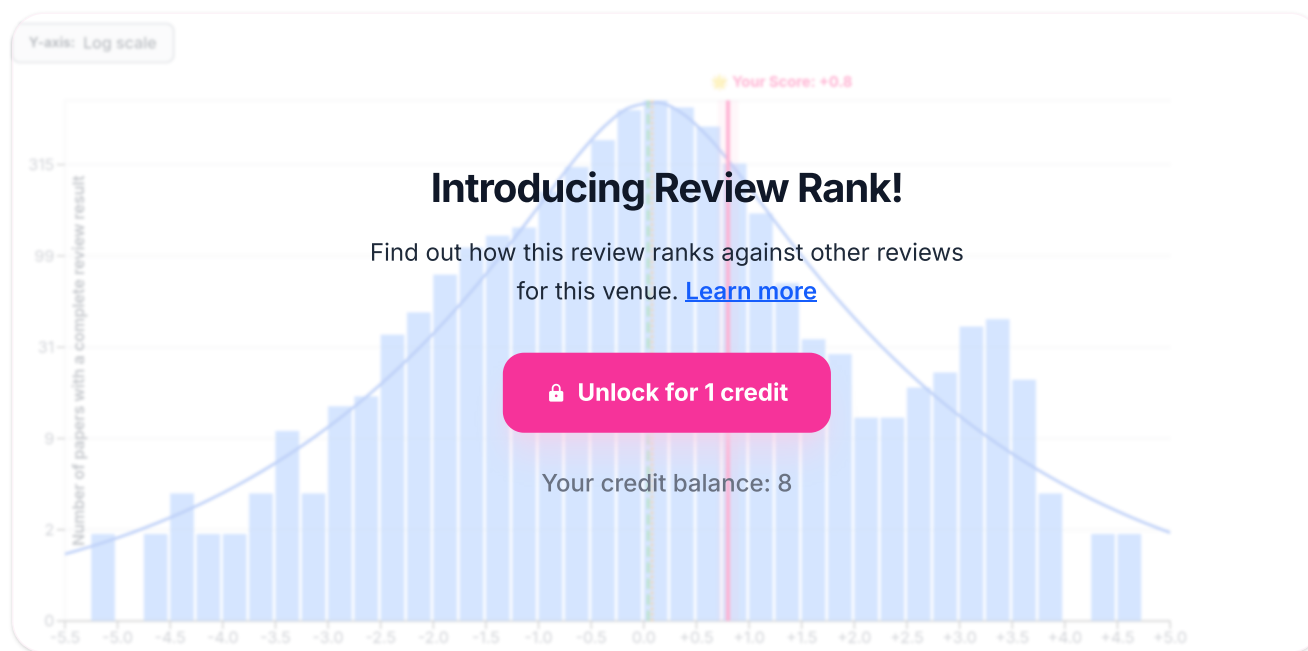


The U-Shaped Challenge of Language Grounding: How Instruction Granularity Induces VLA Shortcuts

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Reviewed as if submitted to CONFERENCE ACL 2026 - main track ⓘ



Desk Rejection Assessment:

Paper Length

Pass .

Topic Compatibility

Pass ☒. The paper studies instruction granularity for language grounding in embodied VLA agents, which squarely fits ACL's Multimodality and Language Grounding to Vision/Robotics and Analysis of NLP Models areas.

Minimum Quality

Pass . The submission contains Abstract, Introduction, Related Work, Method (Sections 3.1–3.4), Benchmark (Section 4), Experiments and Results (Section 5), Conclusion (Section 6), and a Limitations section (Section 7). The work is scientifically framed with quantitative results and figures.

Prompt Injection and Hidden Manipulation Detection

Pass . I found no attempts to manipulate automated reviewing or hidden prompts.

Expected Review Outcome:

Paper Summary

This paper proposes using rule width (w), adapted from width-based planning, as an agent-centric formalism to quantify instruction granularity for language-grounded VLA agents. Instructions are represented as rule sets over symbolic features; the width of the hardest rule, $w(\mathcal{R})$, quantifies the peak number of concurrent state features that must be coordinated to follow the instruction. The authors introduce Mini-BEHAVIOR-Gran, a controlled benchmark derived from Mini-BEHAVIOR with 20 tasks and multiple width levels while keeping perception and action space fixed. Empirically, they report a U-shaped trend in in-distribution success rates (SR) when training and testing on single granularity: high SR for fine ($w=0$), a dip at medium ($w=1$), and a rebound for coarse ($w \geq 2$). They attribute the rebound to increased instruction predictability $P(I|v)$, which encourages shallow, visuomotor policies; mixed-granularity training partly mitigates this. They also observe an asymmetric transfer: coarse-trained models transfer to fine instructions better than the reverse.

Summary Of Strengths

- Clear agent-centric framing of “instruction granularity” that focuses on decision burden rather than textual surface properties. Section 3 formalizes rules over a shared feature space Φ and defines w as the maximum concurrent feature dependency. This is a thoughtful bridge between classical planning insights and neural VLA training.
- The benchmark attempts to control for confounds by holding perception and discrete control constant and varying granularity via solvable rule sets (§4). Figure 2 gives a comprehensive view of the pipeline, including the dynamic rule selection by the oracle instructor and three different action-decoding heads.
- The U-shaped effect is well-documented across three decoding paradigms (autoregressive, discrete diffusion, parallel decoding) in Figure 5, lending credibility that the phenomenon is architectural-agnostic within the study’s scope.
- Figures 3 and 4 help decouple granularity from sequence length: Figure 3 shows that rule calls decrease as instructions become coarser, and Figure 4’s heatmap shows monotonic SR decay with width even when conditioning on instruction horizon, supporting the claim that w captures latent planning difficulty.
- The diagnosis of shallow grounding is supported by two complementary analyses: (a) a perturbation study contrasting full instructions vs static goal language vs no language (Figure 6), and (b) a VLM-based instruction prediction probe showing lower cross-entropy for coarser instructions (Figure 7). Together these align with the $P(I|v)$ hypothesis.
- Table 2 is informative: it shows the generalization asymmetry (train coarse $\rightarrow$ test fine works better than the reverse) and that mixed-granularity training (especially Axial-F) improves overall SR while increasing language sensitivity.

Summary Of Weaknesses

1. Dependence on a hand-designed symbolic feature space and rule templates (Sections 3.2–3.4). While the paper argues that width is agent-centric and transferable, w is only well-defined after choosing Φ and the rule template set. The selection of features and decomposition rules is subjective and may vary across task designers, which could change measured widths and potentially the U-shaped effect. The paper would be stronger with an analysis of robustness to alternative, equally valid Φ choices or rule-template realizations for the same tasks.
2. Oracle instructor and scope of the claim about “grounding demand.” The experiments assume an oracle that selects an available rule r_t and emits its natural-language realization at each step (Figure 2). This reduces the agent’s responsibility to single-step grounding conditioned on the current state and selected rule, sidestepping the open problem of instruction following under imperfect, temporally extended guidance. The claim that “latent planning demand w is intrinsic” (§7) is reasonable, but because the oracle collapses the space of language-to-action mappings, it is hard to generalize these findings to more realistic, non-oracular settings where agents must select among multiple candidate rules or infer missing steps. A control where the oracle only provides the rule set $\mathcal{R}$ (not r_t) would help bound how much the results depend on the oracle.
3. Evidence for the $P(I|v)$ mechanism is suggestive but not conclusive. The explanation of the rebound at coarse granularity relies on a probability-aggregation argument and a VLM classifier’s cross-entropy (Figure 7) trained to predict I from v . However, the mapping from v to I is constructed via deterministic templates of the same rules that also govern action supervision, which risks circularity. Stronger causal evidence would include: (i) controlling $P(I|v)$ directly by shuffling or adversarially perturbing language at training time while keeping width fixed, (ii) estimating mutual information $I(L;V)$ or conditional entropy $H(L|V)$ under matched distributions across widths, or (iii) occluding or noising visual features to test whether the coarse-trained policy indeed over-relies on vision.
4. Confounding between width and task distribution/horizon persists. Figure 4 tries to decouple width from instruction horizon via the Centroid mixture, but the heatmap cells are populated by different underlying tasks and some cells are missing (“nan”), indicating sparse coverage. It remains possible that particular tasks dominate certain width–horizon bins and drive the trend. A stronger control would match identical tasks across widths where feasible, or perform per-task paired comparisons when multiple width realizations exist.
5. Limited external validity and ambiguity at high width. Experiments run exclusively in Mini-BEHAVIOR with discrete actions (§4–§5.1). The strong claim of a “capacity cap” at $w \geq 4$ (Table 2) could alternatively stem from limited exposure to such cases during training, from curriculum issues, or from distributional mismatch across widths. The paper asserts all rule sets are solvable, but it is unclear whether the training schedule provided sufficient examples for $w \geq 4$ or whether those cases are rare. Reporting the frequency of $w \geq 4$ samples in training and testing, and running a targeted curriculum on only $w \geq 4$ would make the “capacity cap” claim more convincing.

Additional, more minor issues:

- The definition of fulfillment of r_i (Section 3.3) assumes “keeping unmentioned features constant.” This is a strong assumption for embodied execution and may not hold in settings with incidental changes; implications for robustness deserve discussion

beyond the brief note on novelty pruning.

- The paper does not compare to a non-learning, width-aware baseline (e.g., BFWS) under the same instruction regimes to probe whether the U-shape is unique to learned policies or a more general property.
- Some notation in Table 1 mixes parentheses/brackets and exhibits typos (e.g., “ $pr = X$ ”, ..., “ $pr = 0$ ”), which hampers readability of the running example.

Figure- and table-specific comments:

- Figure 4 convincingly demonstrates that, holding a coarse measure of horizon (rule calls) roughly fixed, SR still declines as width increases, supporting w as a better complexity surrogate than token-level heuristics. However, the “nan” cells highlight uneven coverage and suggest caution in over-interpreting the strict monotonicity across the entire grid.
- Figure 5’s U-shape is clear across three distinct decoders, strengthening the empirical claim. Still, confidence intervals or variation across seeds would improve interpretability.
- Table 2 is central to the generalization claim and the “capacity cap” conclusion. The complete failure for $Width \geq 4$ under all training regimens is striking; without exposure statistics and a curriculum analysis, it is difficult to separate an intrinsic bottleneck from data scarcity.

Potentially Missing Related Work

1. Pugacheva et al., “Bring the Apple, Not the Sofa: Impact of Irrelevant Context in Embodied AI Commands on VLA Models,” 2025 — Studies how extraneous language affects VLA behavior. Directly relevant to the paper’s analysis of instruction informativeness and shallow grounding. Should be discussed in Related Work (Section 2) and connected to the perturbation analysis in Section 5.4.
2. Li, Zhao, Yang, “Learning Instruction-Guided Manipulation Affordance via Large Models,” 2024 — Explores instruction-conditioned affordances, relevant to how granularity constrains decision spaces. Cite in Section 2 and discuss how width-based granularity might interact with affordance prediction.
3. Gao, Ye, Zhang, “Compressor-VLA: Instruction-Guided Visual Token Compression,” 2025 — Uses language to modulate visual processing. This relates to the paper’s $P(I|V)$ hypothesis; include in Section 2 and consider as an architectural angle in Section 5.1.
4. Nguyen, Suganuma, Okatani, “Look Wide and Interpret Twice,” 2021 — Improves interactive instruction following via multi-pass interpretation, highly pertinent to granularity and grounding. Discuss in Section 2 and contrast with rule-based granularity.
5. Grazioso and Suglia, “An Analysis of Visually Grounded Instructions in Embodied AI Tasks,” 2023 — An analysis paper on visually grounded instructions. Add to Section 2 to better situate the benchmark within prior analytical studies.
6. Körner, Castillo, Drijvers, “Embodied Processing at Six Linguistic Granularity Levels: A Consensus Paper,” 2023 — Theoretical framing of granularity levels; cite in Section 2 to strengthen the conceptual link between linguistic levels and agent-centric width.
7. Huang, Xia, Xiao, “Inner Monologue: Embodied Reasoning through Planning with Language Models,” 2022 — Early work on language-guided planning; discuss in Section 2 and compare with the oracle- r_t setting in Figure 2.
8. Shaikh, Gligoric, Khetan, “Grounding Gaps in Language Model Generations,” 2024 — Investigates grounding gaps, resonant with the paper’s shallow grounding diagnosis. Add to Section 2 and discuss in Section 5.4.
9. Goel, Singh, Choi, “Language Guided Meta-Control for Embodied Instruction Following,” 2025 — Proposes meta-control for better grounding; discuss as an alternative to oracle selection in Figure 2.
10. Wang, Yu, Wu, “Weakly-supervised VLM-guided Partial Contrastive Learning for Visual Language Navigation,” 2024 — Related to instruction-vision interactions under weak supervision. Add to Section 2 and consider as an alternative training signal in §5.5.

Comments Suggestions And Typos

Actionable suggestions:

1. Strengthen the causal case for the $P(I|V)$ mechanism.
 - Control $P(I|V)$ directly: for a fixed w , randomize or adversarially perturb the language surface forms during training to reduce predictability, and measure whether the coarse-regime rebound disappears.
 - Estimate $H(L|V)$ or $I(L;V)$ across widths with matched task distributions to replace the proxy CE loss in Figure 7.
 - Add a vision degradation ablation (e.g., occlusion/noise) to see if coarse-trained policies degrade more, which would further support the visuomotor-shortcut claim.
2. Reduce oracle dependence in at least one setting.
 - Provide a variant where the oracle supplies only $\mathcal{R}$ and not r_t ; the agent must select the applicable rule itself. Report the new U-shape and transfer asymmetry to show robustness.
 - Alternatively, provide a “no-oracle” baseline with subgoal detection learned from experience.
3. Clarify coverage and training exposure for high widths.
 - Report the frequency of samples per width level in training and test sets, and add a curriculum that focuses on $w \geq 4$ to fairly test the “capacity cap.”
 - If feasible, add learning curves stratified by width.
4. Robustness to feature design.
 - Instantiate alternative, reasonable Φ and rule templates for a subset of tasks and show whether width assignments change and whether Figure 5’s U-shape persists.

- Discuss how one might automatically induce Φ and rules from demonstrations, and whether the granularity conclusions are expected to hold.
- 5. Add a planning baseline and statistics.
- Evaluate a width-based planner (e.g., BFWS) directly under the various rule sets to test whether the U-shape or the $w \geq 4$ collapse also appears in symbolic search.
- Provide confidence intervals or standard errors for Figure 5 and Figure 6; currently it is hard to judge the stability of the reported gaps.

Typos and clarity:

- Table 1 uses inconsistent parentheses/brackets (e.g., " $pr = X$), ..., ($pr = 0$]" ; " $Hr, ps = X$), ..., ($Hr, ps = 0$]"). Please fix.
- Section 3.3, the notation for effects and conditions silently assumes frame axioms ("keeping unmentioned features constant"). State this assumption earlier and discuss its impact.
- Page 6, "allow ing" $\rightarrow$ "allowing."
- Ensure that all acronyms (e.g., SR, CE) are defined at first use in the main text.

Criteria that could raise my score:

- A controlled experiment reducing oracle strength (agent chooses r_t) that preserves the U-shape and transfer asymmetry.
- Direct manipulation/measurement of $P(I|v)$ addressing the circularity concern, with corresponding changes in Figure 5/6 patterns.
- Exposure accounting and a curriculum study that either validates or revises the "capacity cap" at $w \geq 4$.

Confidence

4 Quite sure after careful checking: I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness

3.5 Between Acceptable and Strong: The methodology is coherent and analyses are largely supportive, but some claims (predictability mechanism, capacity cap) need stronger causal tests and coverage reporting.

Excitement

3.0 Interesting: The width-based lens and the U-shaped finding are interesting and likely to spur discussion and follow-up work.

Overall Assessment [See how you rank](#)

3.5 Borderline Conference: The paper is close to being acceptable for the main conference but has notable weaknesses. The agent-centric granularity formalization and the careful empirical look at U-shaped performance are valuable. However, the reliance on an oracle selector, incomplete causal evidence for the $P(I|v)$ mechanism, and limited clarity around high-width coverage constrain the strength of the conclusions. With additional controls and analysis, this could become a solid conference paper.

Best Paper Justification

N/A.

Limitations And Societal Impact

The Limitations section acknowledges discrete action spaces and the oracle setup. I recommend also adding a brief societal impact statement: if shallow grounding is incentivized by certain instruction styles, deployed agents might appear to "follow instructions" while ignoring critical language details, which can be unsafe in real-world settings. Discuss mitigations (e.g., mixed-granularity training, counterfactual data, uncertainty reporting). Also consider inclusivity and multilingual aspects: width depends on rule realization; discuss whether templates or Φ choices bias performance across languages or user groups.

Ethical concerns

None.

Needs Ethics Review

No.

Reproducibility

4 Mostly reproducible: The benchmark description, rule-translation templates, and architectural settings are reasonably detailed; releasing the Mini-BEHAVIOR-Gran generation code and instruction templates would make this a 5.

Datasets



3 Potentially useful: If released, Mini-BEHAVIOR-Gran would be helpful for controlled studies of instruction granularity.

Software

3 Potentially useful: Releasing the oracle instructor, rule banks, and training scripts would be useful to others.

Disclaimer: Only the main paper is reviewed; anonymity and formatting are not checked. The desk-rejection assessment evaluates content completeness, correctness, novelty, clarity, and scientific quality in line with common ACL review guidelines.

Reviewed on March 11, 2026



GPT-5

How was this review?

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